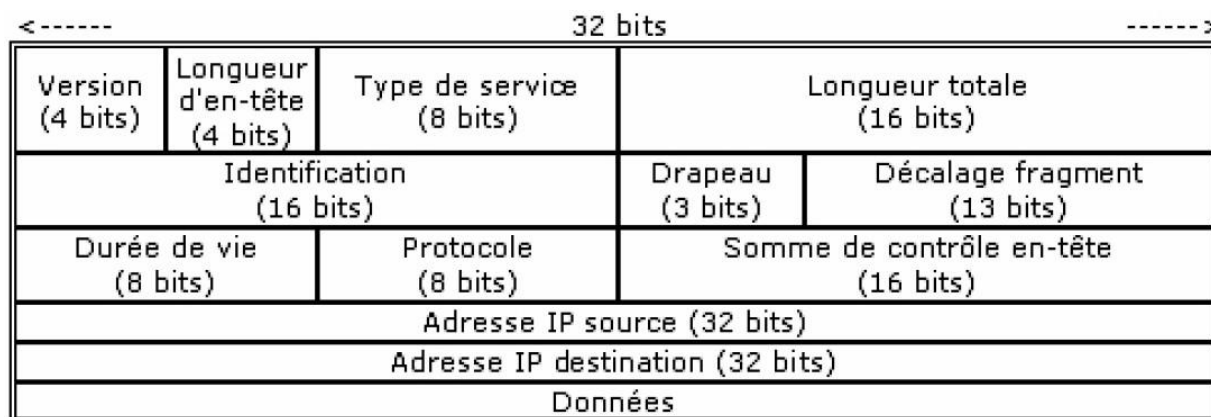


Lab No. 6: Study of Internet layer protocols (TCP/IP)

The Internet layer is very important; it defines the datagrams (data packets of the IP layer). It allows the routing of datagrams to remote machines. This flow of packets (datagrams) is managed by the IP protocol. This layer contains, among others, 3 main protocols (IP, ARP, ICMP).

Exercise 1: The IP Datagram



Here's a clear, structured table describing the **IP Datagram (IPv4)** and its fields:

Field Name	Size (bits)	Description
Version	4	Indicates the IP version. For IPv4, this value is 4.
Header Length (IHL)	4	Specifies the length of the header in 32-bit words.
Type of Service (ToS)	8	Defines packet priority and handling (e.g., delay, throughput, reliability).
Total Length	16	Total size of the IP datagram (header + data) in bytes.
Identification	16	Unique value assigned to identify fragments of the same datagram.
Flags	3	Controls fragmentation: DF (Don't Fragment), MF (More Fragments)
Fragment Offset	13	Indicates the position of a fragment in the original datagram.
Time to Live (TTL)	8	Limits packet lifetime; decremented by each router to prevent infinite loops.
Protocol	8	Identifies the upper-layer protocol (e.g., TCP, UDP, ICMP).
Header Checksum	16	Error-checking for the header only.
Source IP Address	32	IP address of the sender.
Destination IP Address	32	IP address of the receiver.
Data (Payload)	Variable	Actual data being transmitted (e.g., TCP segment, UDP datagram).

The IP datagram is the packet format defined by the **IP** protocol (at the Internet layer).

1. **Start a network capture and collect, in a table (see the figure above), the different information of an IP datagram** (i.e., the values in the 'Internet Protocol' line in the B section of the ETHEREAL window).

Lab No. 6: Study of Internet layer protocols (TCP/IP)

Filter: tcp Expression... Clear Apply Save Filter Filter FilterB2

No.	Time	Source	Destination	Protocol	Length	Info
436	16.566130000	142.250.74.1/0	192.168.1.5	TLSv1.2	1056	Application Data, Application Data
437	16.607608000	192.168.1.5	142.250.74.170	TCP	54	63247 > https [ACK] Seq=1853 Ack=6772 win=64512 Len=0
440	17.015989000	193.194.66.77	192.168.1.5	TCP	1204	[TCP segment of a reassembled PDU]
441	17.015990000	193.194.66.77	192.168.1.5	TCP	1204	[TCP segment of a reassembled PDU]
442	17.015991000	193.194.66.77	192.168.1.5	TCP	1204	[TCP segment of a reassembled PDU]
443	17.015991000	193.194.66.77	192.168.1.5	TCP	1204	[TCP segment of a reassembled PDU]
444	17.016007000	193.194.66.77	192.168.1.5	TCP	1204	[TCP segment of a reassembled PDU]
445	17.016008000	193.194.66.77	192.168.1.5	TCP	1204	[TCP segment of a reassembled PDU]
446	17.016008000	193.194.66.77	192.168.1.5	TCP	1204	[TCP segment of a reassembled PDU]
447	17.016009000	193.194.66.77	192.168.1.5	TCP	1204	[TCP segment of a reassembled PDU]
448	17.016113000	192.168.1.5	193.194.66.77	TLSv1.2	54	63250 > https [ACK] Seq=2541 Ack=12498 win=65536 Len=0
449	17.016340000	193.194.66.77	192.168.1.5	TLSv1.2	1204	Application Data

Internet Protocol Version 4, Src: 193.194.66.77 (193.194.66.77), Dst: 192.168.1.5 (192.168.1.5)

Version: 4
Header length: 20 bytes
Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))
Total Length: 1190
Identification: 0x8b83 (35715)
Flags: 0x00
Fragment offset: 0
Time to live: 52
Protocol: TCP (6)
Header checksum: 0x3112 [correct]
Source: 193.194.66.77 (193.194.66.77)
Destination: 192.168.1.5 (192.168.1.5)
[Source GeoIP: unknown]
[Destination GeoIP: unknown]
Transmission Control Protocol, Src Port: https (443), Dst Port: 63250 (63250), Seq: 12498, Ack: 2541, Len: 1204
[2 Reassembled TCP Segments (2136 bytes): #447(986), #449(1150)]

2. Criticize the different values of a datagram header (Field-by-field critique)

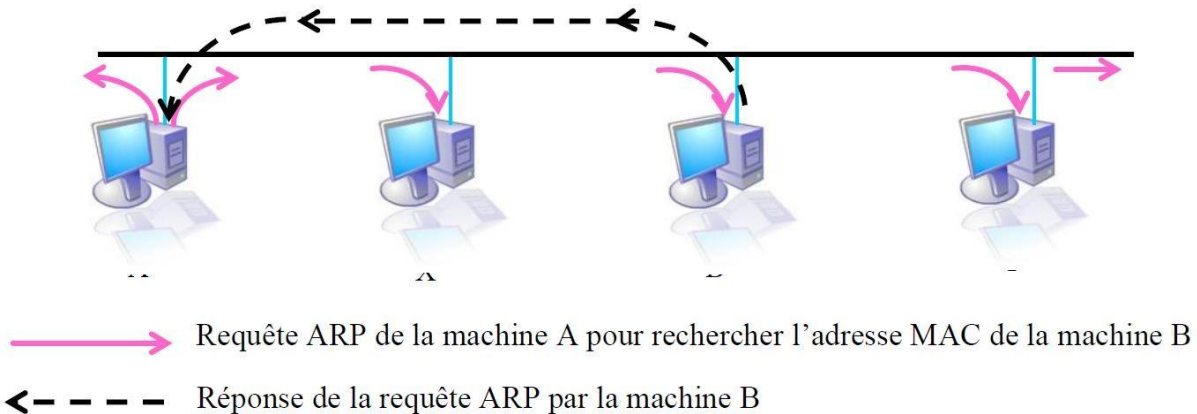
Field	Example Value	Explanation
Version	4	IPv4 protocol, standard. No issue
Header Length (IHL)	20 bytes	Minimum size, meaning no options
Type of Service (ToS)	0x00	Very basic/default (no priority)
Total Length	1190 bytes	Entire packet size, no fragmentation needed MTU (1500 bytes)
Identification	0x8b83 (35715)	Unique ID for fragmentation
Flags	0x00	the packet is not fragmented and is allowed to be fragmented
Fragment Offset	0	Not fragmented
Time To Live (TTL)	52	52 suggests: Packet has traversed ~12 hops (if starting at 64)
Protocol	6 (TCP)	Indicates TCP segment (Standard and expected).
Header Checksum	0x3112 (Correct)	Used for error checking (correct means no issues)
Source IP Address	193.194.66.77	Sender address (Public IP)
Destination IP Address	192.168.1.5	Destination (Private IP)
Options	None	No optional fields
Data (Payload)	TCP segment	Expected (Encapsulated data)

Another example 2: Captured IP Datagram (IPv4)

Field	Example Value	Explanation
Version	4	IPv4 protocol
Header Length (IHL)	24 bytes	Header includes options
Type of Service (ToS)	0x10	Signifies a request for Minimize Delay (or low delay) for network traffic
Total Length	190 bytes	Small packet
Identification	0x5768 (22376)	Used for fragmentation tracking
Flags	0x02	Fragmentation not allowed ("Don't Fragment" is enabled)
Fragment Offset	0	Expected since packet is not fragmented
Time To Live (TTL)	64	Very typical default (Linux/Unix devices)
Protocol	17 (UDP)	Indicates UDP datagram
Header Checksum	0x5f70	Correct
Source IP Address	192.168.1.1	Private IP
Destination IP Address	192.168.1.5	Private IP
Data (Payload)	UDP segment	Could be DNS query

Lab No. 6: Study of Internet layer protocols (TCP/IP)

Exercise 2: The ARP Protocol



It is recalled that communication between machines can only occur through the physical interface. Applications only know IP addresses. To establish the link between IP address/physical address, TCP/IP uses the ARP protocol (Address Resolution Protocol, the working principle of which is explained during the course). For this exercise, we want to study the **ARP table** that holds the IP address/physical address mappings on each machine.

1. To display the content of this table, type the command in MS-DOS: **arp -a**. What is the result?

The command **arp -a** is used to display the **ARP (Address Resolution Protocol) table** on your system.

```
Select Command Prompt
Microsoft Windows [Version 10.0.19043.1237]
(c) Microsoft Corporation. All rights reserved.

C:\Users\peregr>arp -a

Interface: 192.168.69.2 --- 0x8
Internet Address      Physical Address      Type
192.168.69.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static

Interface: 192.168.122.97 --- 0x10
Internet Address      Physical Address      Type
192.168.122.1         52-54-00-86-2b-60    dynamic
192.168.122.255       ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
```

Now, we want to study the contents of the ARP packet. To do this, clear the ARP table of the machines with the command **arp -d**, configure ETHEREAL to capture only ARP packets (in the capture menu/options, then in the capture filter field, enter arp), launch ETHEREAL, and then run a simple ping command from machine **A** to machine **B** on the local network.

2. What is the new content of the ARP table?

If you did something like: `ping 192.168.1.1`
`arp -a`

Lab No. 6: Study of Internet layer protocols (TCP/IP)

For example, you will now see an entry like:

192.168.1.1 aa-bb-cc-dd-ee-ff dynamic

3. Retrieve the different information related to the packets: ARP request and ARP response — that is, the values of the 'Address Resolution Protocol' line in section B of the ETHEREAL window for each selected ARP packet type.

Filter: arp Expression... Clear Apply Save Filter FilterB2

No.	Time	Source	Destination	Protocol	Length	Info
35	12.568912000	IntelCor_e1:19:c8	64:85:05:0c:ff:ec	ARP	42	who has 192.168.1.1? Tell 192.168.1.5
36	12.570723000	64:85:05:0c:ff:ec	IntelCor_e1:19:c8	ARP	42	192.168.1.1 is at 64:85:05:0c:ff:ec
88	79.107483000	64:85:05:0c:ff:ec	IntelCor_e1:19:c8	ARP	42	who has 192.168.1.5? Tell 192.168.1.1

Frame 35: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 1
Ethernet II, Src: IntelCor_e1:19:c8 (84:3a:4b:e1:19:c8), Dst: 64:85:05:0c:ff:ec (64:85:05:0c:ff:ec)
Destination: 64:85:05:0c:ff:ec (64:85:05:0c:ff:ec)
Source: IntelCor_e1:19:c8 (84:3a:4b:e1:19:c8)
Type: ARP (0x0806)
Address Resolution Protocol (request)
Hardware type: Ethernet (1)
Protocol type: IP (0x0800)
Hardware size: 6
Protocol size: 4
Opcode: request (1)
Sender MAC address: IntelCor_e1:19:c8 (84:3a:4b:e1:19:c8)
Sender IP address: 192.168.1.5 (192.168.1.5)
Target MAC address: 64:85:05:0c:ff:ec (64:85:05:0c:ff:ec)
Target IP address: 192.168.1.1 (192.168.1.1)

0000 01100100 10000101 00000101 00001100 11111
0008 01001011 11100001 00011001 11001000 00001
0010 00001000 00000000 00000110 00000100 00001
0018 01001011 11100001 00011001 11001000 11001
0020 01100100 10000101 00000101 00001100 11111
0028 00000001 00000001

Filter: arp Expression... Clear Apply Save Filter FilterB2

No.	Time	Source	Destination	Protocol	Length	Info
35	12.568912000	IntelCor_e1:19:c8	64:85:05:0c:ff:ec	ARP	42	who has 192.168.1.1? Tell 192.168.1.5
36	12.570723000	64:85:05:0c:ff:ec	IntelCor_e1:19:c8	ARP	42	192.168.1.1 is at 64:85:05:0c:ff:ec
88	79.107483000	64:85:05:0c:ff:ec	IntelCor_e1:19:c8	ARP	42	who has 192.168.1.5? Tell 192.168.1.1

Frame 36: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 1
Ethernet II, Src: 64:85:05:0c:ff:ec (64:85:05:0c:ff:ec), Dst: IntelCor_e1:19:c8 (84:3a:4b:e1:19:c8)
Destination: IntelCor_e1:19:c8 (84:3a:4b:e1:19:c8)
Source: 64:85:05:0c:ff:ec (64:85:05:0c:ff:ec)
Type: ARP (0x0806)
Address Resolution Protocol (reply)
Hardware type: Ethernet (1)
Protocol type: IP (0x0800)
Hardware size: 6
Protocol size: 4
Opcode: reply (2)
Sender MAC address: 64:85:05:0c:ff:ec (64:85:05:0c:ff:ec)
Sender IP address: 192.168.1.1 (192.168.1.1)
Target MAC address: IntelCor_e1:19:c8 (84:3a:4b:e1:19:c8)
Target IP address: 192.168.1.5 (192.168.1.5)

0000 10000100 00111010 01001011 11100001 00011
0008 00000101 00001100 11111111 11101100 00001
0010 00001000 00000000 00000110 00000100 00001
0018 00000101 00001100 11111111 11101100 11001
0020 10000100 00111010 01001011 11100001 00011
0028 00000001 00000101

Field	ARP Request	ARP Response	Description
Hardware type	1 (Ethernet)	1 (Ethernet)	Type of hardware used
Protocol type	0x0800 (IPv4)	0x0800 (IPv4)	Type of protocol being mapped
Hardware size	6	6	Length of MAC address in bytes
Protocol size	4	4	Length of IP address in bytes
Opcode	1 (request)	2 (reply)	1 = request, 2 = reply
Sender MAC address	84: 3a:4b:e1:19:c8	64:85:05:0C:FF:EC ^E	MAC of sender (requester / IP owner)
Sender IP address	192.168.1.5	192.168.1.1	IP of sender (requester / IP owner)
Target MAC address	64:85:05:0C:FF:EC	84: 3a:4b:e1:19:c8	MAC being queried (unknown in request, filled in reply)
Target IP address	192.168.1.1	192.168.1.5	IP being queried or target